JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 96, NO. B3, PAGES 4051-4064, MARCH 10, 1991

# MAGNETIC POLARITY STRATIGRAPHY AND CARBONIFEROUS PALEOPOLE POSITIONS FROM THE JOGGINS SECTION, CUMBERLAND STRUCTURAL BASIN, NOVA SCOTIA

V. J. DiVenere[1] and N. D. Opdyke

Department of Geology, University of Florida, Gainesville

Abstract. One hundred seventy-seven samples were taken in stratigraphic sequence in redbeds from approximately 2100 m of Namurian-Westphalian strata from the Shepody, Enragé, Boss Point, and lower Joggins formations between Minudie and Joggins, Nova Scotia, for the purpose of constructing a magnetic polarity stratigraphy. In addition, five samples from each of four sites were taken from north dipping strata of the Cumberland Group for the purpose of conducting a fold test for the Joggins Formation. This yields a total coverage of 412 samples from approximately 2400 m of south dipping strata in the composite section when combined with previous results from the Maringouin Peninsula (DiVenere and Opdyke, 1990) and 88 samples from north-dipping strata collected for the purpose of conducting fold tests. Two components of magnetization are identified in these sediments: a prefolding component, termed C, and a synfolding component, termed B. The present results extend the mid-Carboniferous magnetic polarity stratigraphy to include eight normal and eight reversed magnetozones in the Namurian A through early Westphalian A interval. The results are dominantly to completely reversed in samples from the remainder of the Westphalian A and the Westphalian B, but because of the complication of varying degrees of remagnetization and an 800 m sampling gap it is not possible to verify the base of the Permo-Carboniferous reversed superchron in this time period. Northward plate motion is inferred from an observed shallowing of inclinations upsection. A 5-7 cm/yr average poleward drift rate is calculated for this time period from these results, yielding a low-latitude southern hemisphere paleogeographic position.

## Introduction

DiVenere and Opdyke [1990] have recently demonstrated that red sediments of Namurian age from the Maringouin and Shepody formations exposed on the Maringouin Peninsula in New Brunswick yield an internally coherent magnetic polarity stratigraphy. The purpose of this study is to extend the magnetic polarity stratigraphy of the Maringouin Peninsula study to younger sediments across Cumberland Basin in Nova Scotia and attempt to determine the lower boundary of the Permo-Carboniferous Reversed Superchron (PCRS). The base of the PCRS is an important time line [Roy and Morris, 1983] and in combination with a developed magnetic polarity stratigraphy for the time preceding could be an important tool for intercontinental stratigraphic correlations.

Paleomagnetic work by Roy and others in red sandstones of the Cumberland structural basin of New Brunswick and Nova Scotia has shown the possibility that the base of the PCRS could be identified in these sediments. Roy and Park [1969] found mixed polarities in the early Westphalian A Enragé Formation of the Maringouin Peninsula in New Brunswick. Roy [1969] observed only reversed polarity in the Westphalian B Cumberland Group Facies A (hereafter called the Joggins Formation as mapped by the Nova Scotia Department of Mines and Energy). It was concluded that the Joggins Formation must have been deposited during the PCRS.

## Geologic Setting

Good exposures of Carboniferous red sediments of Namurian and Westphalian age are found in coastal outcrops on the south dipping limb of the Minudie Anticline along the eastern shore of Cumberland Basin in western Nova Scotia (Figure 1). The Joggins section contains 4000 m of Upper Carboniferous strata and is well known for its terrestrial vertebrate fossils and in situ trees. The Shepody, Claremont, and Boss Point formations of western Nova Scotia are equivalent to the Shepody, Enragé, and Boss Point formations on the Maringouin Peninsula of western New Brunswick. The Minudie Anticline in Nova Scotia appears to be an extension of the Maringouin Anticline on the Maringouin Peninsula.

Although the Claremont/Boss Point and the Boss Point/Joggins contacts are conformable, previous workers considered the contact between the Shepody and Enragé/Claremont formations to be unconformable. This was based on interpretation of fossil megaplants and nonmarine invertebrates [Bell, 1944] which seemed to indicate that the Namurian B and C were missing from the sequence. However, Belt [1964] considered the evidence for an unconformity to be incomplete based on a later study of the type Namurian section in Belgium by Stockmans and Willière [1953]. Copeland [1957], studying arthropods, found a gradation of Namurian to possibly Westphalian A age with only slight faunal variation and no apparent lithologic break between the Shepody and Claremont formations. Belt [1965] believed that the Claremont Conglomerate conformably overlies the Shepody Formation in the Joggins section of western Nova Scotia, although there is an unconformity between equivalent units in the Minas Basin in southern Nova Scotia. No evidence for an unconformity was seen in the field during this study. The sampled sequence from the Shepody Formation through the Joggins Formation is considered conformable. The stratigraphic section sampled in the Cumberland structural basin is summarized in Figure 2.

### Shepody Formation

The Shepody Formation is a coarsening upward fluvial sequence of red mudstone, siltstone, and sandstone interbedded with green/gray sandstone and matrix supported conglomerates. Belt [1965] extracted microspores of Barss's microspore zone c, which is considered to be Namurian C, from the upper Shepody Formation at Downings Cove, on Cumberland Basin.

### Claremont Conglomerate

The Claremont Conglomerate is composed of red to purple sandstone and conglomerate with interbedded siltstone. The conglomerates are generally matrix supported with angular to subrounded quartz shards, lithic fragments, and red limeclasts in a sand matrix. The age of the formation is believed by most workers [Bell, 1944; Belt, 1965] to be Westphalian A.

[1] Now at Lamont-Doherty Geological Observatory and Department of Geological Sciences, Columbia University, Palisades, New York.

Copyright 1991 by the American Geophysical Union.

Paper number 90JB02148
0148-0227/91/90JB-02148$05.00

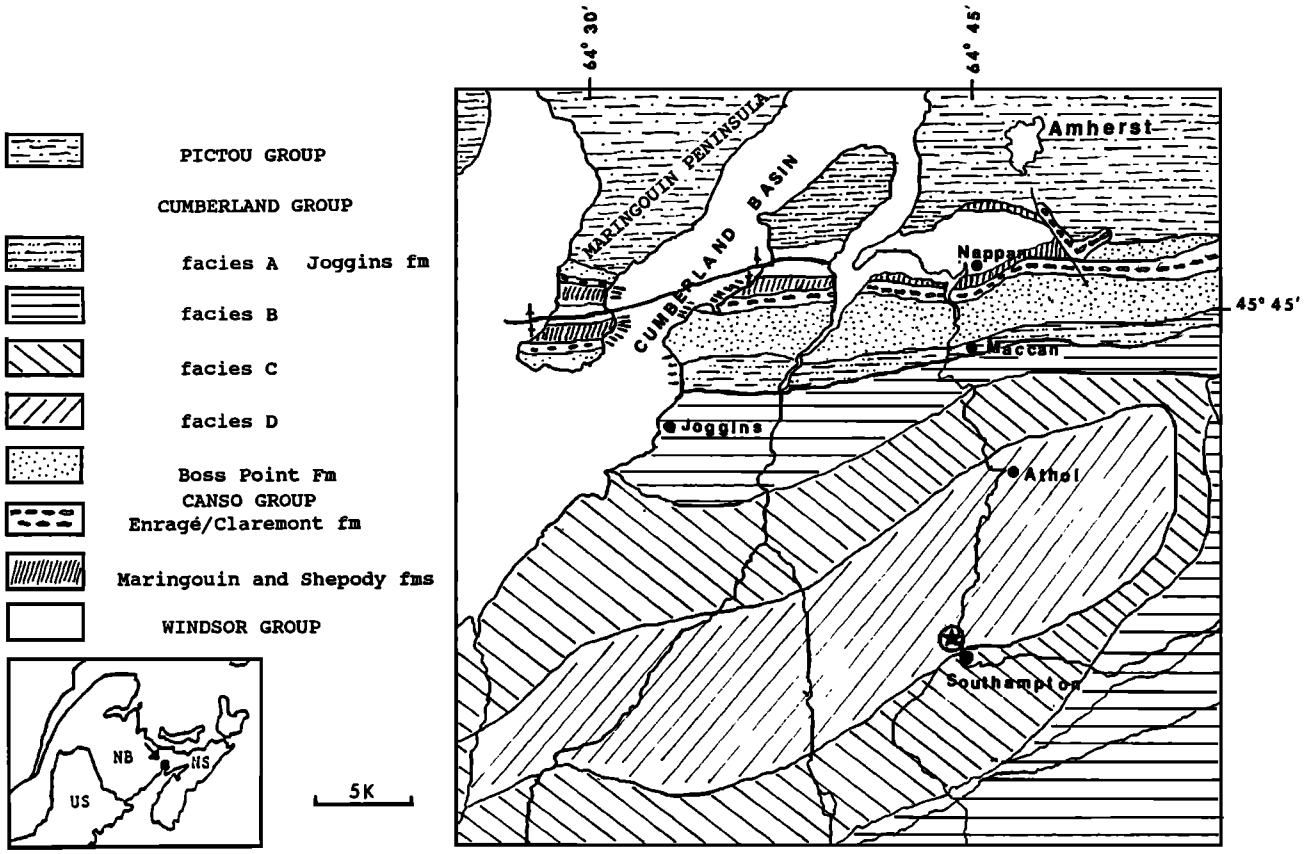



Fig. 1. Cumberland structural basin sampling area. Hatched shorelines represent continuously sampled section. Star is sampling site of north dipping Cumberland Group.

## Boss Point Formation

The base of the Boss Point Formation as mapped by the Nova Scotia Department of Mines and Energy (1988) on the shore of Cumberland Basin consists of approximately 50 meters of red mudstone, siltstone, and sandstone. These finer sediments found in Boss Cove are considered to be early to mid-Westphalian A [Nova Scotia Department of Mines and Energy, 1988]. Stratigraphically above this is 800 m of green, crossbedded sandstone with plant material and minor conglomerates. Plant material includes *Calamites* debris and in situ tree stumps. The upper 180 m of the Boss Point Formation is composed of red mudstone, siltstone, and sandstone interbedded with green sandstone. The Boss Point Formation above the basal red facies is placed as mid to late Westphalian A [Nova Scotia Department of Mines and Energy, 1988].

## Joggins Formation

The lower part of the Joggins Formation, equivalent to Facies A of the Cumberland Group of previous workers [Bell, 1944; Belt, 1965] is composed of red shales interbedded with red to gray sheet sandstones conformably overlying the Boss Point Formation. Plant fossils and pelecypods found in the gray shales of the upper Joggins Formation (Cumberland Group Facies B) are absent [Carroll et al., 1972]. The lower Joggins Formation does not form coastal cliffs as does the upper Joggins Formation or the underlying Boss Point Formation but is exposed in discontinuous shoreline outcrops in Lower Cove. Bell [1944] judged by the flora that these strata were no older than Westphalian A and no younger than early Westphalian B; he preferred an early Westphalian B age. Copeland [1957] used arthropods to confirm a Westphalian B age. The Nova Scotia Department of Mines and Energy [1988] considers the units overlying the Joggins Formation to be early to mid-Westphalian B in age.

## Sampling and Sample Treatment

One hundred seventy-seven samples were taken in stratigraphic sequence in redbeds from approximately 2100 m of Carboniferous strata from the Shepody, Enragé, Boss Point, and lower Joggins formations. The samples were taken from the south dipping limb of the Minudie Anticline in western Nova Scotia on the shore of Cumberland Basin and Chignecto Bay between Minudie and Joggins. Single samples were taken at each sampled horizon. In addition, five oriented cores from each of four sites were taken in north dipping strata of the Joggins Formation at River Maccan. The sampling area and geology are shown in Figure 1.

A portable, gasoline-powered drill was used to obtain the samples. Cores were oriented with a Brunton compass mounted on an orienting device. Strike and dip measurements were taken throughout the sequence. The 2.25 cm diameter cylindrical samples were later cut to 1.25-1.75 cm lengths. Section thickness was measured by tape for closely spaced samples and a combination of pace and Brunton and eye-height and Brunton where sampling was discontinuous.

The sequence was sampled beginning in the Shepody Formation north of Downings Cove. Twenty-four oriented cores were taken from approximately 290 m of strata from the Shepody Formation reference section in Nova Scotia in order to correlate this section with the New Brunswick section across the bay by means of magnetostratigraphy. Thirty

21562202b, 1991, B3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/90JB02148 by Curtin University Library, Wiley Online Library on [14/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

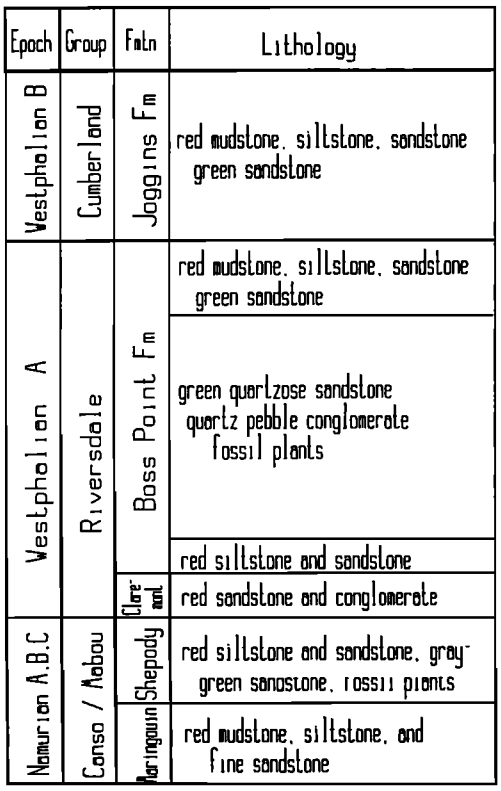



Fig. 2. Stratigraphic section sampled in the Cumberland structural basin of Nova Scotia and New Brunswick.

oriented cores were taken in stratigraphic sequence from approximately 130 m of the Claremont Conglomerate. Sampling density was limited in the Claremont Conglomerate by the spacing of the finer grained-beds. Twelve oriented cores were taken sequentially from approximately 50 m of the lower red facies of the Boss Point Formation. Three oriented hand samples from each of two sites, one red siltstone and one gray shale, were taken from the 800 m of green sandstone of the Boss Point Formation on Boss Point. Forty-three oriented cores were taken in stratigraphic sequence from approximately 180 m of the upper red facies of the Boss Point Formation. One hundred fifty meters of section were covered or contained no redbeds at the boundary between the Boss Point Formation and the Joggins Formation. Sixty-two oriented cores were taken sequentially from approximately 475 m of the lower Joggins Formation in Lower Cove, north of the town of Joggins.

Demagnetization was performed in a Schonstedt TSD-1 thermal demagnetizer. The remanent magnetization of the samples was measured in an SCT two-axis cryogenic magnetometer at the University of Florida beginning with the NRM. Storage and all procedures (other than cutting) were performed in a double-walled transformer steel room [Scott and Frohlich, 1985] with an average field of 50-250 nT. In addition, the thermal demagnetizer and magnetometer are housed inside nested mu-metal cylinders, and samples were kept inside a single mu-metal cylinder after treatment and before measurement.

After the natural remanent magnetization (NRM) was measured samples were incrementally heated in 100°C steps to 600°C. Above 600°C the samples were treated in 10°C steps to 660°C and 5°C steps thereafter. Some samples were treated to a maximum temperature of 705°C.

## Results

### Demagnetization Behavior

NRM. Typically, the measured intensity of the NRM is of the order $10^{-3}$ to $10^{-2}$ A/m. Sample NRM declinations are almost entirely reversed. Only two samples in the Shepody Formation and one sample in the Claremont Conglomerate display normal NRM.

A component. Many samples show some evidence at low temperatures of a weak component of magnetization which might be a viscous remanent magnetization acquired in the recent field. However, it is too weak to be resolved and is normally destroyed by heating to 100°-200°C. This viscous tail is clearly seen in Figures 3a and 3b.

B component. Beginning at 100°-300°C, a component of magnetization is seen in the orthogonal plots which decays with a linear trajectory to temperatures above 600°C. It is southerly directed and typically has positive inclination. Long, linear segments at intermediate temperatures are seen in the orthogonal plots from the Claremont Conglomerate (Figures 3a-3c).

C component. Both normal and reversed polarity are observed at high temperature with maximum blocking temperatures in the 685°-690°C range. Inclinations are typically shallow negative for the normal directions and shallow positive for reversed samples.

The Shepody Formation results are not as stable as those from the Maringouin Peninsula section across Cumberland Basin due to the coarser nature of the sampled outcrop (the finer-grained beds tended to be well weathered or covered). Nevertheless, polarities are clear, with reversed polarity at the bottom and top of the formation and a large band of normal in the middle as seen in the Maringouin Peninsula Results. A normal sample from the Shepody Formation is shown in Figure 3a.

Most Claremont Conglomerate samples show a decay trajectory at intermediate temperatures toward a northwest and shallow up direction but either bunch up about an endpoint at high temperatures without a linear decay to the origin, indicating a nondistributed population of magnetic grains stable at high temperatures, or unblock at an intermediate direction (often at 270°), indicating overlapping unblocking temperature spectra of a reversed B component and a normal C component.

A family of decay curves is recognized, all interpreted as samples with normal polarity C components. A sample with clearly normal polarity but no linear decay to the origin is shown in Figure 3a. Another sample is seen to move to the north but destabilize after crossing the east-west axis in Figure 3b. A third sample, less stable at high temperatures (or with a greater degree of overlap of B and C components), moves toward a normal endpoint but unblocks as it reaches the east-west axis (Figure 3c). The last point on the north axis is probably not significant.

At the highest temperatures (over 675°C) the decay may be generally toward the origin or it may be erratic. Twenty-one Claremont Conglomerate samples display normal high-temperature directions or a decay toward this direction. The other seven samples remain reversed without necessarily yielding a linear decay to the origin. Figures 3b and 3c show the demagnetization behavior of two samples from the Claremont Conglomerate.

The finer sediments of the Boss Point Formation in Boss Cove yield stable high temperature results. Nine of the 12 samples display reversed directions with linear decay to the origin at temperatures above 610°C and a maximum unblocking temperature between 680°C and 690°C. A reversed sample in the lower Boss Point Formation is shown in Figure 3d. Two consecutive samples display trajectories toward northerly endpoints but unblock at intermediate declinations as in Figures 3b and 3c. Above the 800 m of green sandstone of Boss Point, almost all samples display reversed high temperature directions. Only two samples in the upper red facies of the Boss Point Formation and three samples in the

21562202b, 1991, B3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/90JB02148 by Curtin University Library, Wiley Online Library on [14/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

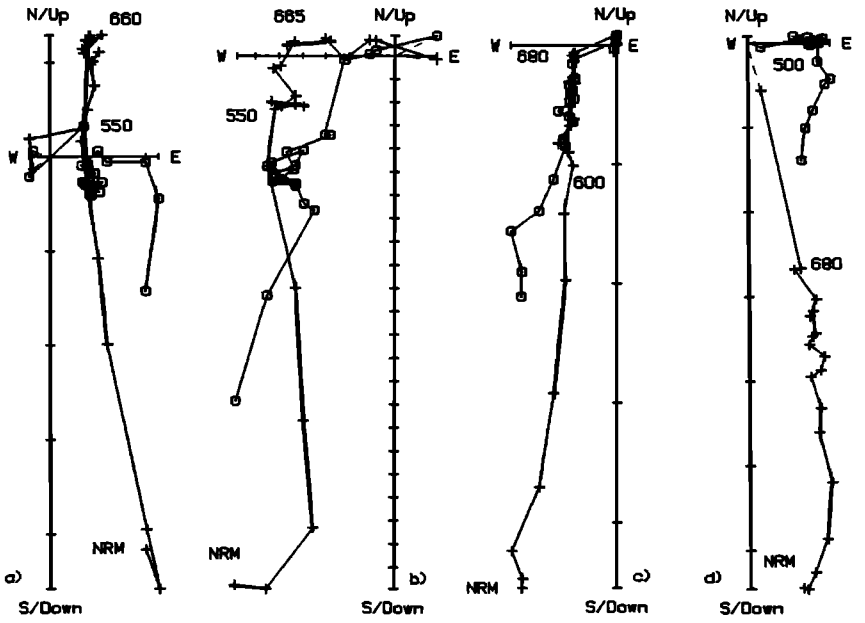



Fig. 3. Orthogonal plots in Figures 3a and 3b are from the Claremont Conglomerate and those in Figures 3c and 3d are from the lower Boss Point Formation. Units are $10^{-3}$ A/m for Figures 3a, 3c, and 3d and $10^{-4}$ A/m for Figure 3b. The vertical axes are north and up. A clearly normal sample is shown in Figure 3a, apparently normal in Figure 3b, an unresolved sample in Figure 3c moving toward a normal endpoint, and a clearly reversed sample with bivectoral decay in Figure 3d.

Joggins Formation show any indication of normal polarity. Maximum blocking temperatures are in the 680°-690°C range. There is typically little or no directional change observed between intermediate and high temperatures in samples from the Boss Point Formation upper red facies. Nearly complete univectoral behavior is seen in Figure 4a, a sample from the upper red facies of the Boss Point Formation. However, most samples from the Joggins Formation do show some directional change, as shown in Figure 4b where the inclination moves to a constant shallow-up value and the declination swings in toward the origin at high temperature. A sample from the Joggins Formation is shown in Figure 4c which appears to have an unresolved normal component of magnetization.

## Statistical Treatment

Principal components [Kirschvink, 1980] were calculated for each sample on linear segments of data chosen by eye from orthogonal plots. High-temperature (C) components which decayed toward the origin were anchored to the origin. Unresolved normal samples were fitted with great circles [McFadden and McElhinny, 1988]. The principal components (both B and C) were then subjected to standard Fisher statistics [Fisher, 1953] or McFadden and McElhinny's [1988] method for combining direct observations and great circles to calculate mean declinations, mean inclinations, the precision parameter (k), and $\alpha_{95}$.

## Mean Directions

Shepody Formation. The C component from the Shepody Formation on the Maringouin Peninsula in New Brunswick was shown to have a positive fold test and reversal test [DiVenere and Opdyke, 1990]. A reversal test was performed for the Nova Scotia Shepody Formation C component by comparing the mean direction calculated from the normal samples with the mean from samples with reversed C components. This is shown in Figure 5a with the reversed mean inverted to the northern hemisphere. The normal

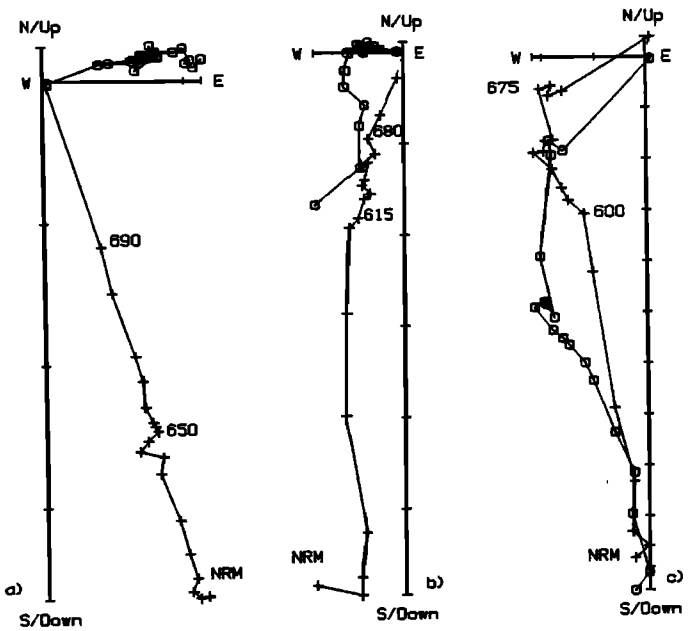



Fig. 4. Orthogonal plots from Lower Cove. Vertical axes are north and up. Units are $10^{-2}$ A/m for Figure 4a and $10^{-3}$ A/m for Figures 4b and 4c. Univectoral decay is seen in (a) a sample from the upper Boss Point Formation (b) clearly multi-component decay in a reversed sample, and (c) an unresolved possibly normal sample in the lower Joggins Formation.

mean direction after tilt correction is D=353.9°, I=-12.4°, and $\alpha_{95}$=9.5° calculated from six direct observations and nine great circles with sector constraints [McFadden and McElhinny, 1988]. The reversed mean is D=184.9°, I=27.2°, and $\alpha_{95}$=25.9 for

21562202b, 1991, B3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/90JB02148 by Curtin University Library, Wiley Online Library on [14/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

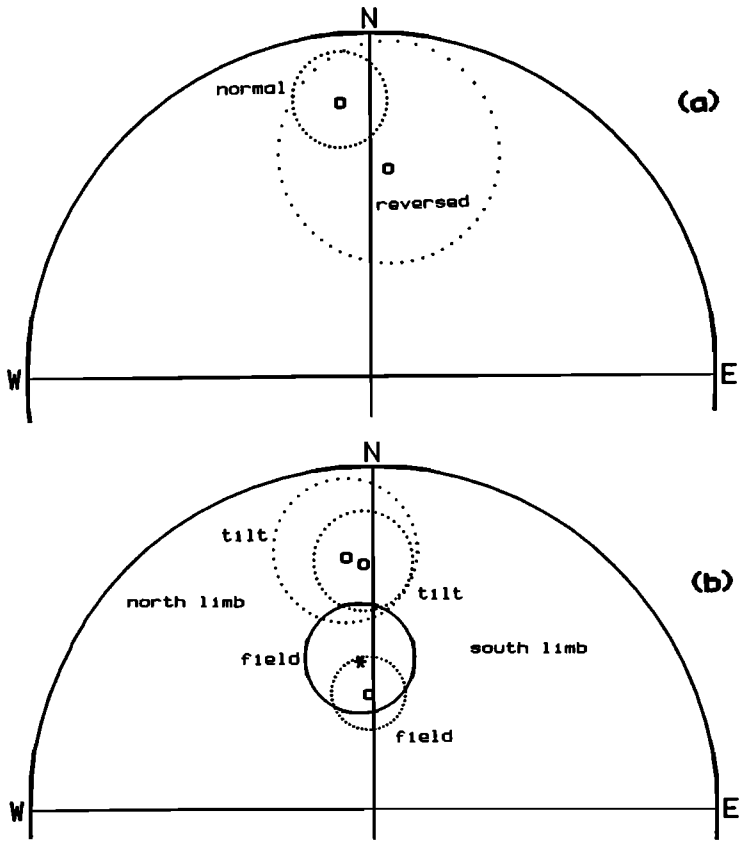



Fig. 5. Shepody Formation (a) reversal test: normal and reversed C component mean directions plotted on a common hemisphere with $\alpha_{95}$ circles of confidence; (b) fold test: north and south limb C component means plotted before (field) and after (tilt) tilt correction. Open symbols and dotted circles are negative inclinations. Stereographic projection.

only eight direct observations. The directions are not significantly different. The larger data set from the Shepody Formation on the Maringouin Peninsula yielded a very good agreement between the normal and reversed means [DiVenere and Opdyke, 1990]. Results from the south dipping limb in Nova Scotia may be compared with the results from the north dipping limb of the Shepody Formation on the Maringouin Peninsula. This comparison yields a positive fold test for the C component (Figure 5b and Table 1). Therefore, it may be concluded that the C component from the Shepody Formation in Nova Scotia was acquired before folding. The mean direction for the C component after tilt correction for the Shepody Formation in Nova Scotia is D=357.9, I=-18.7, and $\alpha_{95}$=10.8 calculated from 14 direct observations and nine great circles with sector constraints. This is in good agreement with the New Brunswick result of D=352.8°, I=-15.6°, and $\alpha_{95}$=4.6° for N=80 [DiVenere and Opdyke, 1990].

The B component in the Shepody Formation of the New Brunswick section was shown to be synfolding with a maximum value of k at 71% of unfolding and D=173.1°, I=19.4°, and $\alpha_{95}$=4.4° for N=58 [DiVenere and Opdyke, 1990]. An incremental fold test was performed for the Nova Scotia results by combining 30 direct observations from the north dipping limb in New Brunswick with 18 direct observations (straight-line segments) observed in the Nova Scotia samples. A maximum value of k is observed at 73% of unfolding with D=176.2°, I=23.3°, and $\alpha_{95}$=5.1° for N=48 (Figure 6 and Table 2).

Claremont Conglomerate. The C component from the Claremont Conglomerate yields a tilt corrected mean of D=353.8°, I=-9.0°, and $\alpha_{95}$=5.8° calculated from one direct observation and 20 great circles. Ten samples of the coeval Enragé Formation were collected from the north dipping limb of the Maringouin Anticline on the Maringouin Peninsula in New Brunswick directly above the Shepody Formation collection. These coarse sandstones yield a tilt corrected C component mean of D=317.2°, I=-3.3, and $\alpha_{95}$=17.4° calculated from nine great circles with sector constraints. The mean declination is unlike that observed in any other older or younger mid-Carboniferous formation in the Cumberland structural basin and probably reflects some difficulty in estimating the true high-temperature component mean direction in the present circumstance with few samples and coarse lithology. The Claremont and Enragé C component means are compared before and after tilt correction in Figure 7. Although the 95% circles of confidence do not overlap at any point during unfolding, it is seen that the north limb mean moves from north and steeply down to north and shallow, while the south limb mean moves from north and steeply up to north and shallow during tilt correction. This is also the case for the Shepody Formation of Nova Scotia as well as both the Maringouin and Shepody formations in the New Brunswick study. Also, the Claremont Formation tilt-corrected C component mean is similar to the Shepody Formation tilt-corrected mean (already shown to have been acquired before folding) (Table 1). Therefore, although the fold test here is not statistically significant, it appears likely that the Claremont C component was also acquired before folding.

An incremental fold test was performed by combining 27 direct observations of the B component from the south dipping beds in Nova Scotia with 10 direct observations from the north dipping beds in

21562202b, 1991, B3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/90JB02148 by Curtin University Library, Wiley Online Library on [14/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1. C Component

| Unit | Dip | Coord | D,deg | I,deg | k | a95,deg | Pole | | N |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | Longitude | Latitude | |
| Shep | S | Field | 357.6 | -52.5 | 8.6 | 11.0 | 117.6 | 11.2 | 14do, 9gc |
| | | Tilt | 357.9 | -18.7 | 8.7 | 10.8 | 118.1 | 34.6 | |
| Shep | N/S | Field | 356.4 | -9.2 | 2.4 | 18.8 | 120.2 | 39.5 | 30do, 9gc |
| | | Tilt | 356.2 | -17.9 | 8.0 | 8.6 | 120.1 | 35.0 | |
| Shep | N | Field | 355.1 | 42.8 | 6.9 | 15.2 | 127.9 | 68.8 | 16do |
| | | Tilt | 354.1 | -17.0 | 6.9 | 15.2 | 122.7 | 35.3 | |
| Clar | S | Field | 352.7 | -44.6 | 37.3 | 5.4 | 126.2 | 46.5 | 1do, 20gc |
| | | Tilt | 353.8 | -9.0 | 32.7 | 5.8 | 123.6 | 39.4 | |
| Enr | N | Field | 297.6 | 56.8 | 12.0 | 17.3 | 218.4 | 43.8 | 9gc |
| | | Tilt | 317.2 | -3.3 | 12.0 | 17.4 | 166.8 | 29.4 | |
| lBoss | S | Field | 176.2 | 46.1 | 28.7 | 8.3 | 119.1 | 16.7 | 10do, 2gc |
| | | Tilt | 176.1 | 9.1 | 28.7 | 8.3 | 120.6 | 39.6 | |
| Jog | S | Field | 175.6 | 41.6 | 15.8 | 4.8 | 119.8 | 20.2 | 60do |
| | | Tilt | 176.2 | 13.8 | 16.0 | 4.7 | 120.3 | 37.2 | |
| Jog | N/S | Field | 173.7 | 32.9 | 10.9 | 5.0 | 122.2 | 26.1 | 80do |
| | | 85% | 174.2 | 17.4 | 16.9 | 4.0 | 122.6 | 35.1 | |
| | | Tilt | 174.2 | 14.6 | 16.8 | 4.0 | 122.7 | 36.6 | |
| Jog | N | Field | 169.5 | 6.6 | 21.9 | 7.1 | 129.3 | 40.1 | 20do |
| | | Tilt | 168.1 | 17.1 | 22.1 | 7.1 | 129.9 | 34.5 | |
| | | Field | 169.8 | 6.6 | 99.6 | 9.3 | 128.9 | 40.1 | 4 sites |
| | | Tilt | 168.3 | 17.2 | 101.6 | 9.2 | 129.6 | 34.5 | |
| Joga | S | Field | 182.6 | 32.4 | 24.6 | 4.8 | 112.8 | 26.6 | 38do |
| | | Tilt | 181.9 | 4.8 | 24.7 | 4.8 | 113.0 | 41.8 | |

Unit: Shep, Shepody; Clar, Claremont; Enr, Enrage; lBoss, lower Boss Point; Jog, Joggins; Joga, Joggins anchored; Dip: N, north dipping beds; S, south dipping beds; N/S, combined; Coord: field coordinates or tilt corrected; D, mean declination; I, mean inclination; k, Fisher's precision parameter; a95, radius of cone of 95% confidence interval; Pole: mean paleomagnetic pole longitude latitude; N, number of results: do, direct observations; gc, great circles; sites.

New Brunswick. The maximum value of k is observed at 77% of unfolding (Figure 6b and Table 2) with a combined mean D=176.4°, I=21.4°, and $\alpha_{95}$=3.9° for N=37.

Boss Point Formation. The magnetic results from the Boss Point Formation are divided into two groups, the lower red facies and the upper red facies because they are separated by 800 m of green sandstone from which only one site yields useful results. The lower red facies yields a tilt-corrected mean direction for the C component of D=176.1°, I=9.1°, and $\alpha_{95}$=8.3° calculated from 10 direct observations from samples with reversed C components combined with two great circles with sector constraints from two consecutive normal samples. This is very nearly the same direction as the Claremont mean C component after tilt correction. The B component mean direction after tilt correction is D=180.4°, I=12.9°, and $\alpha_{95}$=2.9 for N=12. The B component mean was recalculated at 75% of unfolding for comparison with B components with the other formations which have fold tests. The B component mean at 75% of unfolding is D=180.6°, I=20.6°, and $\alpha_{95}$=3.0° for N=12 (Table 2).

The gray site displayed reversed polarity and contained a high-coercivity component which was not demagnetized by 100 mT but was unstable during thermal demagnetization. The red site within the green sandstone facies yields all reversed polarity and a shallow mean inclination (after tilt correction) with high scatter. These results were not used in calculation of any mean directions reported here.

The upper red facies yields a mean direction after tilt correction of D=174.8°, I=14.4°, and $\alpha_{95}$=4.6° for N=36. This direction is similar to the tilt corrected means for the B components seen in the south dipping beds of the Cumberland structural basin. The inclination is perhaps steeper than expected for a C component at this stratigraphic position (see following discussion of shallowing inclination trend). Although a few samples show small trends to lower inclination and two samples, as previously mentioned, move toward a normal endpoint, it appears that, overall, these samples are largely remagnetized by the B component and any C component remaining in these sediments is too weak to be resolved by these methods.

Joggins Formation. Most samples from the Joggins Formation show a shallowing of inclination at high temperatures, though in many cases it is only slight. The tilt-corrected mean direction for the south dipping Nova Scotia beds calculated from unanchored high temperature segments is D=176.2°, I=13.8°, and $\alpha_{95}$=4.7° for N=60. The tilt corrected mean for the north dipping beds from River Maccan is D=168.1°, I=17.1°, and $\alpha_{95}$=7.1° for N=20 samples (D=168.3°, I=17.2°, $\alpha_{95}$=9.2°, N=4 calculated by site). The distribution of sample directions is shown in Figure 8 for the north and south-dipping beds combined both before and after tilt correction and at 85% of tilt correction. The mean directions appear to cross during unfolding. Incremental unfolding was performed (Figure 6c) and the results show a plateau of precision (k) values above 85% of unfolding with just a slight drop near 100% of unfolding. It should be noted that the north dipping River Maccan sites are not in the lower Joggins Formation but are farther up in the Cumberland Group sequence. This may account for the plateau of k values.

An incremental fold test was performed for the B component by combining 40 sample directions from the south dipping Joggins beds with 20 sample directions from the north dipping River Maccan beds (Figure 6d). The maximum value of k is obtained at 66% of unfolding with D=169.9°, I=16.9°, and $\alpha_{95}$=4.3° for N=60.

## Magnetic Stratigraphy

Normal and reversed magnetozones are observed including normal polarity in sediments of Westphalian A age. The magnetostratigraphy from the

21562202b, 1991, B3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/90JB02148 by Curtin University Library, Wiley Online Library on [14/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

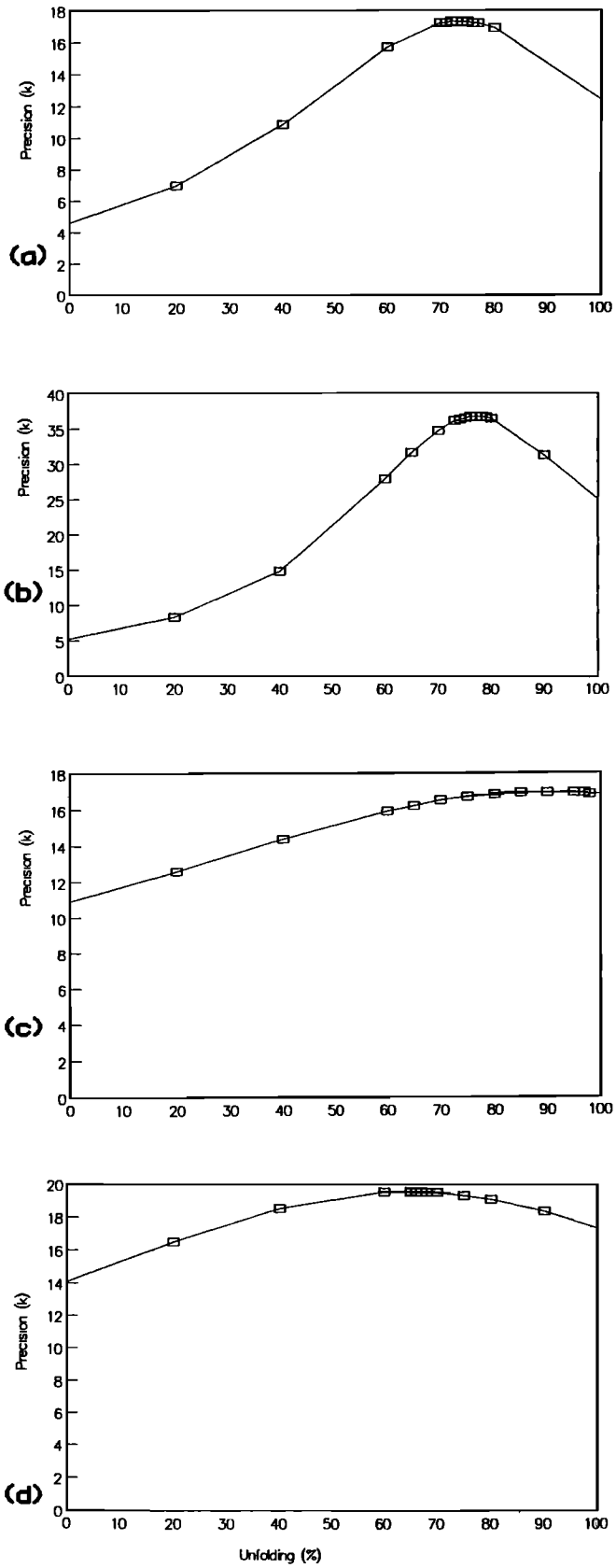



Fig. 6. Incremental fold tests plotting Fisher's precision parameter (k) vs percent of unfolding: (a) Shepody Formation B component, (b) Enragé/ Claremont B component, (c) Joggins Formation C component, (d) Joggins Formation B component.

Shepody, Claremont, and lower Boss Point formations is shown in Figure 9 plotting normalized VGP latitude versus stratigraphic position. VGP latitude was normalized by calculating the angular distance of sample VGP to its respective formation mean pole (i.e., VGP latitude is calculated in relation to the ancient pole, not the modern pole). Crosses represent samples which were analyzed via great circles (samples with incompletely resolved normal C components). Directions were assigned to each great circle by calculating the nearest point on the great circle to the respective formation mean.

The Shepody Formation has a large normal polarity zone in mid-formation with reversed polarity below and near the top. Normal polarity is seen near the Shepody/Claremont contact. The pattern is essentially the same as that seen in the results from the Maringouin Peninsula [DiVenere and Opdyke, 1990].

Magnetozones are more difficult to determine in the Claremont Conglomerate due to the coarse nature of the sediments and poor high temperature stability. However, close inspection of the demagnetization behavior permits a reasonable interpretation. As described above, many samples are clearly normal. Several others unblock at intermediate positions along paths toward fully normal directions. A few samples show no evidence for normal polarity, but some of these do not decay toward the origin and probably contain a normal component as well. On this basis the Claremont Conglomerate may be divided into four magnetozones: two normal and two reversed. The formation is reversed at its contact with the Shepody Formation based on the lowest 2 samples and normal near its upper contact with the lower red facies of the Boss Point Formation. A single reversed sample with no sign of a normal component occurs within a normal polarity zone. No polarity zone is ascribed to this single result.

The lower red facies of the Boss Point Formation is dominantly reversed. However, two consecutive samples show decay trajectories toward a normal endpoint before unblocking at an intermediate direction. These samples are considered normal.

The one red site within the green sandstones of the Boss Point Formation yields reversed results. There is a 110 m gap between the Boss Cove sediments and this site. There is another 590 m interval between this site and the upper red facies.

Of the 86 samples taken during the first field season in the upper red facies of the Boss Point Formation and the lower Joggins Formation most display completely reversed behavior (Figure 10). However, the demagnetization behavior of five nonadjacent samples indicates the possibility of an unresolved normal component (one displays an intermediate direction and four display trajectories moving toward full normal endpoints). This could be evidence for either short-lived normal polarity magnetozones or a large degree of remagnetization obscuring larger normal polarity magnetozones.

An additional 19 oriented cores were obtained from the lower Joggins Formation in Lower Cove at the 1950-1980, 2060, 2135, and 2260 m levels in the composite section (Figure 10) in order to verify the presence of normal polarity at these levels (as inferred from samples with trajectories moving toward normal endpoints). All samples were subjected to progressive thermal demagnetization up to a temperature of 690°C. No evidence was seen to confirm normal magnetozones.

A positive correlation may be made between the Nova Scotia section and the Maringouin Peninsula section (Figure 10) based on the three magnetozones in the Shepody Formation and a distinctive lithologic change from the Shepody Formation to the Claremont Conglomerate of Nova Scotia/Enragé Formation of New Brunswick. The lower three magnetozones of the Nova Scotia section correlate to R4, N5, and R5 of the Maringouin section

21562202b, 1991, B3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/90JB02148 by Curtin University Library, Wiley Online Library on [14/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 2. B Component

| Unit | Dip | Coord | D,deg | I,deg | k | a95,deg | Pole | | N |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | Longitude | Latitude | |
| Shep | S | Field | 168.6 | 47.8 | 24.6 | 7.3 | 125.9 | 14.7 | 17 |
| | | Tilt | 172.0 | 14.8 | 23.7 | 7.5 | 125.4 | 36.3 | |
| Shep | N/S | Field | 176.0 | 2.1 | 4.6 | 10.8 | 121.0 | 43.1 | 48 |
| | | 73% | 176.2 | 23.3 | 17.3 | 5.1 | 119.9 | 32.0 | |
| | | Tilt | 177.8 | 30.4 | 12.4 | 6.1 | 117.9 | 27.9 | |
| Shep | N | Field | 178.9 | -21.7 | 14.9 | 6.9 | 117.5 | 55.5 | 31 |
| | | Tilt | 181.6 | 35.5 | 14.9 | 6.9 | 113.9 | 24.6 | |
| Clar | S | Field. | 178.9 | 49.1 | 44.2 | 4.2 | 116.5 | 14.3 | 27 |
| | | Tilt | 178.0 | 13.1 | 41.3 | 4.4 | 118.1 | 37.6 | |
| EnCl | N/S | Field | 176.7 | 30.9 | 5.2 | 11.5 | 119.1 | 27.5 | 37 |
| | | 77% | 176.4 | 21.4 | 36.7 | 3.9 | 119.8 | 33.1 | |
| | | Tilt | 176.9 | 18.6 | 25.0 | 4.8 | 119.3 | 34.7 | |
| lBoss | S | Field | 182.6 | 49.7 | 215.6 | 3.0 | 113.3 | 13.7 | 12 |
| | | 75% | 180.6 | 20.6 | 215.5 | 3.0 | 114.9 | 33.6 | |
| | | Tilt | 180.4 | 12.9 | 218.2 | 2.9 | 115.1 | 37.7 | |
| uBoss | S | Field | 174.6 | 43.1 | 27.5 | 4.6 | 120.7 | 19.0 | 36 |
| | | 75% | 174.6 | 21.5 | 27.8 | 4.6 | 121.9 | 32.9 | |
| | | Tilt | 174.8 | 14.4 | 27.7 | 4.6 | 122.0 | 36.8 | |
| Jog | S | Field | 169.6 | 34.6 | 16.4 | 5.8 | 126.4 | 24.6 | 40 |
| | | Tilt | 171.1 | 7.2 | 16.7 | 5.7 | 127.2 | 40.0 | |
| Jog | N/S | Field | 169.5 | 26.5 | 14.1 | 5.1 | 127.3 | 29.5 | 60 |
| | | 66% | 169.9 | 16.9 | 19.6 | 4.3 | 127.8 | 34.9 | |
| | | Tilt | 169.8 | 11.8 | 17.3 | 4.5 | 128.4 | 37.5 | |
| Jog | N | Field | 169.3 | 10.6 | 27.3 | 6.4 | 129.1 | 38.0 | 20 |
| | | Tilt | 167.3 | 20.9 | 28.0 | 6.3 | 130.4 | 32.3 | |

Unit: Shep, Shepody; Clar, Claremont; EnCl, combined Enrage/Claremont; lBoss, lower Boss Point; uBoss, upper Boss Point; Jog, Joggins; Dip: N, north dipping beds; S, south dipping beds; N/S, combined; Coord: field coordinates or tilt corrected; D, mean declination; I, mean inclination; k, Fisher's precision parameter; a95, radius of cone of 95% confidence interval; Pole, mean paleomagnetic pole longitude latitude; N, number of samples.

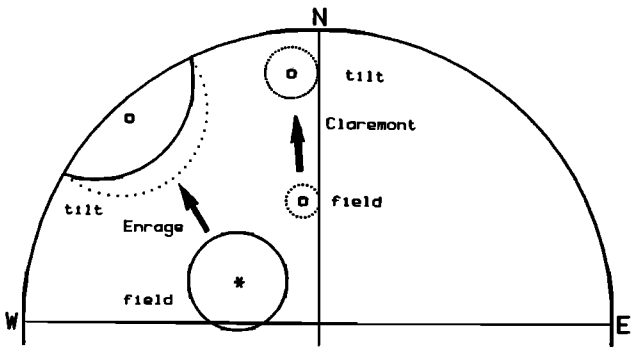



Fig. 7. Enragé/Claremont C component fold test with the north limb (Enragé) and south limb (Claremont) mean directions with $\alpha_{95}$ circles of confidence plotted before (field) and after (tilt) correcting for tilt. Open symbols and dotted circles are negative inclinations. Stereographic projection.

[DiVenere and Opdyke, 1990]. This terminology is continued to R8 in the lower Boss Point Formation.

## Discussion

The stability of the C component of magnetization has been demonstrated previously for the Maringouin and Shepody formations on the Maringouin Peninsula by positive reversal and fold tests [DiVenere and Opdyke, 1990] as well as in the present study for the Shepody Formation of Nova Scotia. Therefore the time of remanence acquisition in these Namurian strata is constrained to be before the time of folding which occurred between the deposition of the upper Joggins Formation (early Westphalian B) and the overlying Pictouan sediments which are Westphalian C and D in age [Bradley, 1982]. The internally consistent magnetic reversal stratigraphy observed, including the matching pattern across Cumberland Basin in the Shepody formation should be further proof of this early magnetization.

The Claremont/Enragé fold test, although not conclusive due to the difficulty of separating the components, appears to indicate a prefolding acquisition for the C component in these Westphalian A strata.

Roy [1969] obtained a positive fold test in the Cumberland Group (Joggins Formation). His south dipping sites were primarily in the lower Joggins Formation at Lower Cove as in the present study, and his north dipping sites were farther up in the Cumberland Group near the River Maccan sites of the present study. Roy generally treated samples to only 550°C, and he calculated mean directions from single temperature steps. His tilt corrected mean is D=172°, I=16°, and $\alpha_{95}$=5° calculated from 10 site means. This may be compared with the present tilt corrected result of D=176.2°, I=13.8°, and $\alpha_{95}$=4.7° for N=60 from south dipping beds at Joggins or the combined tilt-corrected mean D=174.2°, I=14.6°, and $\alpha_{95}$=4.0° for N=80 including 20 samples from the north dipping River Maccan sites. However, the plateau in k values observed during incremental unfolding of the combined data set suggests that either there was significant continental (or at least basinal) motion between the time of deposition of the lower and upper Cumberland Group or

21562202b, 1991, B3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/90JB02148 by Curtin University Library, Wiley Online Library on [14/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

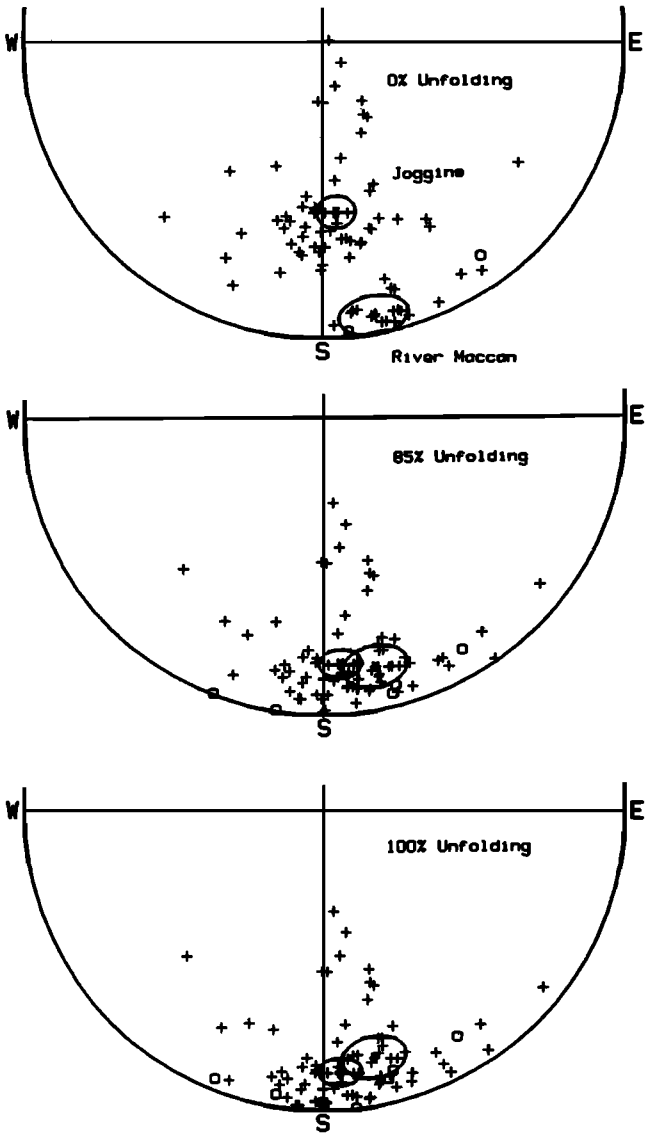



Fig. 8. Joggins Formation C component with the north and south limbs combined at 0%, 85%, and 100% of tilt correction, plotting sample directions along with mean directions from the respective limbs (south dipping: Joggins, north dipping: River Maccan). Open symbols are negative inclinations. Equal-area projection.

that the C component calculated here is contaminated by the B component. In fact, if one recalculates the mean using the same component data from the south limb but anchored to the origin the mean direction from the south limb is shallow: D=181.9°, I=4.8°, and $\alpha_{95}$=4.8° for N=38 after tilt correction. This difference between the anchored and unanchored component means may indicate that the most stable component is not completely resolved at high temperature but is still moving to lower inclinations when the blocking temperature is reached. Therefore Roy's [1969] Cumberland Group pole, which has been used in apparent polar wander paths [Irving and Irving, 1982], may not be a completely reliable estimate of the magnetic field for this time period. The anchored mean for the Joggins formation, though not strictly correct if the inclinations are still moving, is perhaps a better estimate of the field at this time than Roy's [1969] and the unanchored mean described above.

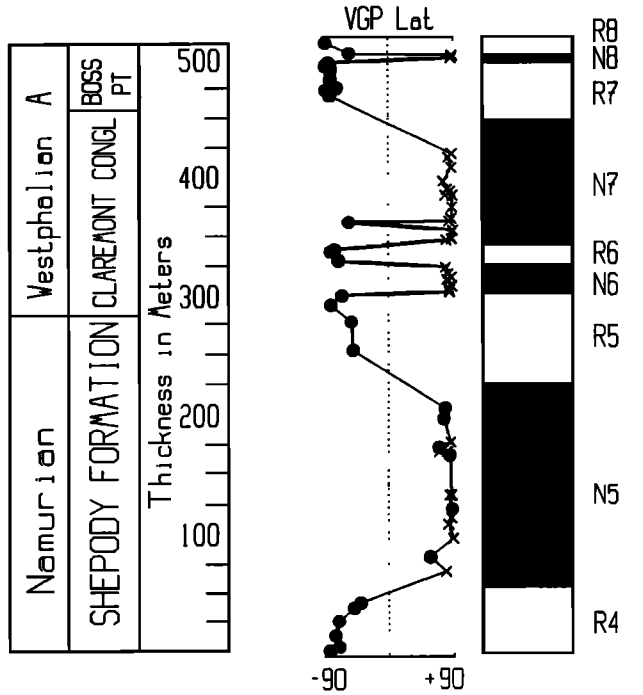



Fig. 9. Magnetostratigraphic results plotting normalized VGP latitude versus stratigraphic position. VGP latitude was normalized by calculating the angular distance of each sample VGP to its respective formation mean pole. Crosses represent samples which were analyzed via great circles (samples with incompletely resolved normal C components). Directions were assigned to each great circle by calculating the nearest point on the great circle to the respective formation mean.

## The Motion of North America

Roy and his colleagues working in the Canadian Maritimes have recorded a systematic change in mean directions with age progression in Carboniferous formations [Roy, 1969; Roy and Morris, 1983]. A progressive shallowing of inclinations has been observed in the present study in the Cumberland structural basin.

In most paleomagnetic studies, pole positions are expressed in terms of formation means. In the present study, single samples were taken at spaced intervals through the section. Since the section on the Maringouin Peninsula is long and continuous and the fine red sediments there yield stable high-temperature results, it was decided to subdivide the section into 100 m intervals in order to study in detail the observed shallowing of inclination up section.

The variation of inclination versus mean stratigraphic position is shown in Figure 11 for 100 m means from the Maringouin Peninsula section, as well as mean inclinations in the Claremont Conglomerate, and the lower red facies of the Boss Point Formation from the Nova Scotia section. The inclinations vary from -26° in the lowest 100 m sampled in the Maringouin Formation to -10.2° in the upper 75 m of the Shepody Formation. This trend is also seen in the Nova Scotia section, though the lower sampling density and poorer high-temperature stability due to the poorer nature of the outcrop prevented the use of 100 m increments. The formation mean inclination for the 290 m sampled in the Shepody Formation south of Minudie is -18.7°. The Claremont Conglomerate (Clare) yields -9.0°. The 50 m of red sediments in the lower Boss Point Formation (lBoss) yield a similar 9.1° inclination.

21562202b, 1991, B3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/90JB02148 by Curtin University Library, Wiley Online Library on [14/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

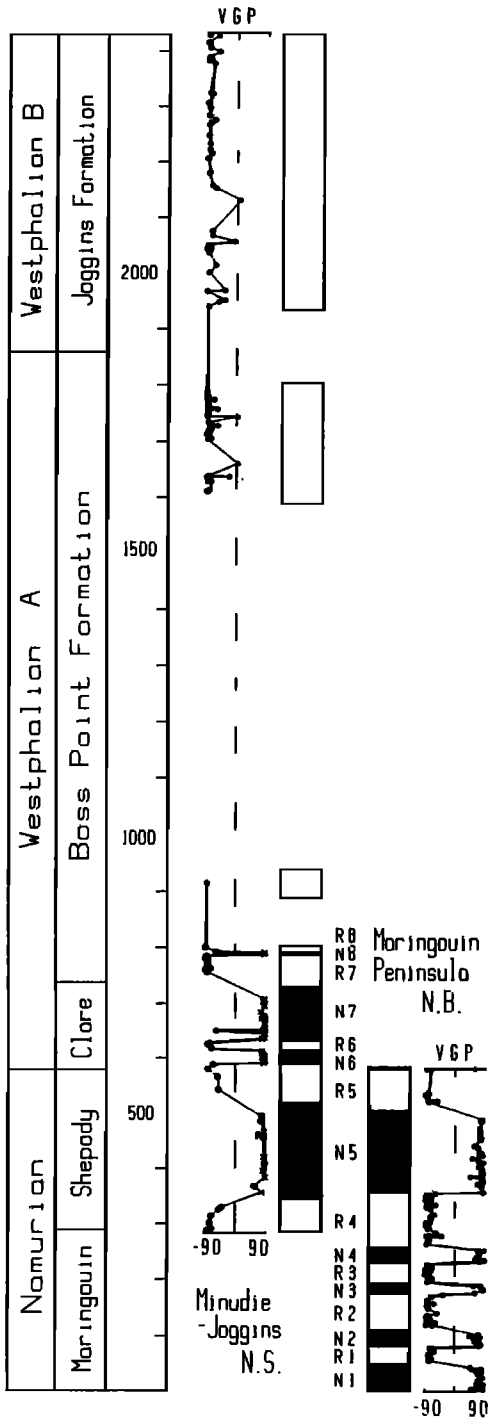



Fig. 10. Magnetostratigraphic results from the Cumberland basin plotting normalized VGP latitude (see text or Figure 10 caption) versus stratigraphic position. Minudie-Joggins, Nova Scotia, are the present results. Maringouin Peninsula, New Brunswick, results are from DiVenere and Opdyke [1990].

There are 700 meters of unsampled green sandstone stratigraphically above the lower red facies of the Boss Point Formation. The upper red facies of the Boss Point Formation (uBoss) is apparently largely overprinted with a mean direction at high temperature which is essentially the same as that of the intermediate temperature B component seen

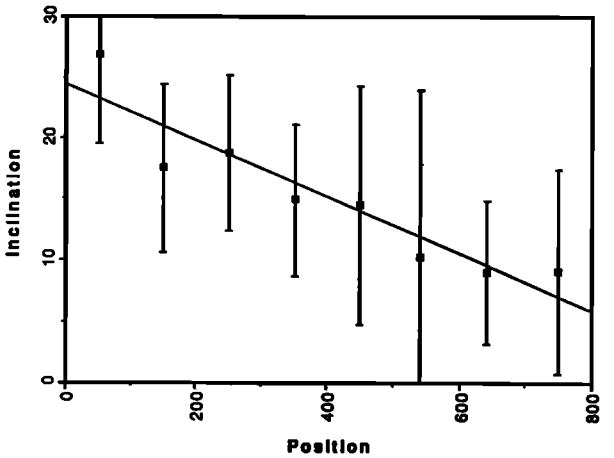



Fig. 11. Variation in inclination with stratigraphic position: stratigraphically lowest 6 data points are from the Maringouin and Shepody formations from the Maringouin Peninsula; the last two points are the Claremont and lower Boss Point mean inclinations from the Nova Scotia section. Error bars are 95% confidence intervals.

throughout the basin. The lower Joggins Formation (Jog), with nearly 500 sampled meters, also appears to be partially overprinted and yields ambiguous results. Its unanchored mean inclination from the south limb is 13.8° and its anchored high-temperature mean (Joga) is 4.8°.

The observed shallowing of inclination from 27° in the lowermost Maringouin Formation to 9° in the Enragé and Boss Point formations indicates a change in paleolatitude of 9.7°, from 14.2°S to 4.5°S, for the period Namurian A through early Westphalian A. This can be compared with the previous estimate of 8° by Roy and Morris [1983] for the Namurian A through Stephanian. The present result is equivalent to 1075 km of poleward motion. The time of deposition is on the order of 15 Ma using the time scale of Harland et al. [1982]. This would yield an average northward drift rate of approximately 7 cm/yr. The measured stratigraphic thickness for this interval is 800 m. This would have resulted from an average sediment accumulation rate over 15 Ma of 5.3 cm/1000 years.

An alternate assessment of poleward drift may be made utilizing the unanchored Joggins C component mean. This would indicate a latitudinal shift of 11.8° (1300 km) from 14.2°S to 2.4°S during the interval Namurian A to Westphalian B which is on the order of 25 m.y. [Harland et al., 1982]. This would yield an average poleward drift rate of approximately 5 cm/yr and an average sediment accumulation rate of approximately 9.6 cm/1000 years for the 2400 m measured in the composite section.

Paleomagnetic poles are plotted in Figure 12 for the Maringouin (Ma) and Shepody (Sh) formations from the Maringouin Peninsula (DiVenere and Opdyke, 1990) and the Shepody Formation (ShN), Claremont Conglomerate (C), Boss Point Formation lower red facies (lBP), and the lower Joggins Formation (Ja, anchored, Ju, unanchored) from Nova Scotia (these results).

Other recently derived North American paleopoles are illustrated in Figure 12 and Table 3. Included is the Carboniferous portion of the polar wander track of Irving and Irving [1982]. Results of McCabe et al. [1989] from the mid-Mississippian Pride Mountain Formation limestones in Alabama (PM) are not significantly different from the results of the present study and from the uppermost Mississippian Mauch Chunk Formation (MC) of Pennsylvania [Kent and Opdyke, 1985; Kent, 1988]. The paleopole obtained from the Deer Lake Formation (DL) of Visean age from Newfoundland [Irving and Strong,

21562202b, 1991, B3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/90JB02148 by Curtin University Library, Wiley Online Library on [14/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

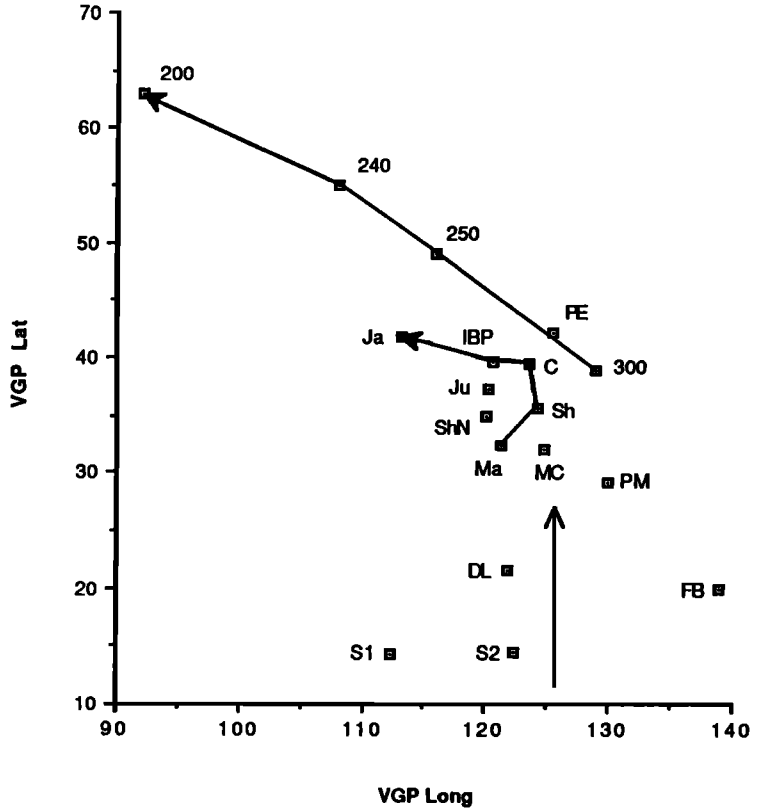



Fig. 12. Upper Paleozoic poles: Ja, Joggins Formation/anchored; Ju, Joggins Formation/unanchored; lBP, lower Boss Point Formation; C, Claremont Conglomerate; ShN, Shepody Formation (these results); Sh, Shepody Formation; Ma, Maringouin Formation [DiVenere and Opdyke, 1990]; PE, Prince Edward Island upper Pictou Group [Symons, 1990]; MC, Mauch Chunk Formation [Kent and Opdyke, 1985]; PM, Pride Mountain Formation [McCabe et al., 1989]; DL, Deer Lake Formation [Irving and Strong, 1984]; FB, Fisset Brook Formation [Johnson and Van der Voo, 1989]; S1 and S2, Scottish poles [Torsvik et al., 1989] rotated into the North American coordinate system using the pole of rotation of Olivet et al. [1984]; 200, 240, 250, and 300 Ma mean poles [Irving and Irving, 1982].

TABLE 3. Upper Paleozoic Poles

| Formation | Age | Symbol | Longitude deg | Latitude deg | a95 deg | Source |
|---|---|---|---|---|---|---|
| Princ Ed Is | early Permian | PE | 125.5 | 42.1 | | Symons [1990] |
| Joggins | Westphalian B | Ja | 113.0 | 41.8 | 4.8 | these results |
| Joggins | Westphalian B | Ju | 120.3 | 37.2 | 4.7 | these results |
| Boss Point | Westphalian A | lBP | 120.6 | 39.6 | 8.3 | these results |
| Claremont | Westphalian A | C | 123.6 | 39.4 | 5.8 | these results |
| Shepody | Namurian | ShN | 120.1 | 35.0 | 8.6 | these results |
| Shepody | Namurian | Sh | 124.3 | 35.7 | 4.6 | DiVenere and Opdyke [1990] |
| Maringouin | Namurian A | Ma | 121.8 | 32.4 | 3.8 | DiVenere and Opdyke [1990] |
| Mauch Chunk | upper Chest | MC | 124.8 | 32.0 | 7.9 | Kent and Opdyke [1985] |
| Pride Mtn | mid Miss | PM | 130.0 | 29.2 | 15.4 | McCabe et al. [1989] |
| Deer Lake | Visean | DL | 121.8 | 21.5 | 8.9 | Irving and Strong [1984] |
| Fisset Brk | Tourn - Namur | FB | 139.0 | 20.0 | 5.1 | Johnson and Van der Voo [1989] |
| Scotland 1 | lower Carb | S1 | 122.3 | 14.5 | 7.0 | Torsvik et al. [1989] * |
| Scotland 2 | lower Carb | S2 | 112.3 | 14.4 | 7.6 | Torsvik et al. [1989] * |
| 300 Ma | | 300 | 129 | 39 | 3 | Irving & Irving [1982] |
| 250 Ma | | 250 | 116 | 49 | 6 | Irving & Irving [1982] |
| 240 Ma | | 240 | 108 | 55 | 6 | Irving & Irving [1982] |
| 200 Ma | | 200 | 92 | 63 | 4 | Irving & Irving [1982] |

* using the pole of rotation of Olivet et al. [1984].

21562202b, 1991, B3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/90JB02148 by Curtin University Library, Wiley Online Library on [14/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

1984] is seen to fall at essentially the same longitude as the Cumberland structural basin results but at a lower latitude by 20°. Synfolding results from the Fisset Brook (FB) Formation of Cape Breton Island [Johnson and Van der Voo, 1989], thought to date from the late Tournaisian through lower Namurian interval, fall at a similar latitude to that of the Deer Lake pole but differ in longitude by 18°. Two lower Carboniferous poles from Scotland (S1 and S2) [Torsvik et al., 1989] when rotated into the North American coordinate system using the pole of rotation of Olivet et al. [1984] fall near 14.5°N latitude.

There is a trend (Figure 12) with paleopole positions moving northward along 125°E longitude from 20°N latitude in Visean times (DL) to approximately 40°N latitude by early Westphalian A time (C, BP). After this the trend moves off toward the northwest along the trend of upper Carboniferous and Permian paleopoles (as is also seen in the Claremont to lower Boss Point portion of the Cumberland structural basin results).

There is the suggestion of a rotation moving from the Shepody pole through the Claremont pole to the lower Boss Point pole. The Joggins poles (Ja and Ju) may be partially overprinted as discussed above. This is approximately the time of the collision with Gondwana and the arc in the paleopole path seen in these data could be the result of basinal rotation during the early stages of this collision. Also included is an early Permian result from Prince Edward Island, Canada of Symons [1990] (PE) with good indications of primary remanence. It falls on the apparent polar wander (APW) path at an older age than expected suggesting that the Carboniferous APW might follow more southerly circuit than that of Irving and Irving [1982].

It is interesting that this change in trend appears to coincide with the approximate onset of the Permo-Carboniferous reversed superchron in Westphalian time. This is perhaps evidence that the onset of the magnetic quiet interval and the final closure of the Proto-Atlantic and the formation of Pangea in Westphalian A time are related. It might be noted that the subduction of an entire ocean basin was being completed at this time. Other implied relationships between tectonics and reversal frequency have previously been noted by several authors [Irving and Pulliah, 1976; Khramov, 1987; Courtillot and Besse, 1987].

## Structural Inferences From Magnetic Results

The relationship of the synfolding B component cannot be easily reconciled with the prefolding C component seen in the Cumberland structural basin results. The B component yields a steeper mean inclination than the C component (Table 1) after applying the appropriate tilt corrections (100% for C, 68% for Maringouin B, 71% for Shepody B (Maringouin Peninsula), 73% for Shepody B (Nova Scotia), 77% for Enragé/Claremont) (Table 2). The B component is also steeper than the C component in situ. The opposite would be expected for the case of continuous northward drift in which the later (B) component would be acquired at lower latitude.

An easy explanation at first glance would be that the B component is the more ancient remanence in spite of its lower temperature stability range. But it is single polarity and almost the same direction in all formations, as opposed to dual polarity and displaying temporal variation as in the C component. Another suggested explanation might be that the B component appears to be synfolding but is actually a prefolding component which has been modified by strain induced rotation during folding. However, if this is the case then the C component would also have been rotated with the B component. But the Maringouin Formation and two Shepody Formation C component fold tests show very close agreement of the north and south limb means after tilt correction and do not support this.

A model can be constructed utilizing the field relationship of the vectors to attempt to resolve the inclination problem. The B component, in situ, has a shallower mean inclination than the C component on the south limb of the Maringouin anticline but is steeper than the C component on the north dipping limb. Model I (Figure 13) shows this relationship. The C component is acquired at time 1 in flat lying beds. At time 2 during folding, the C component has been rotated to a steeper inclination on the south limb and to a shallower inclination on the north limb. The B component is

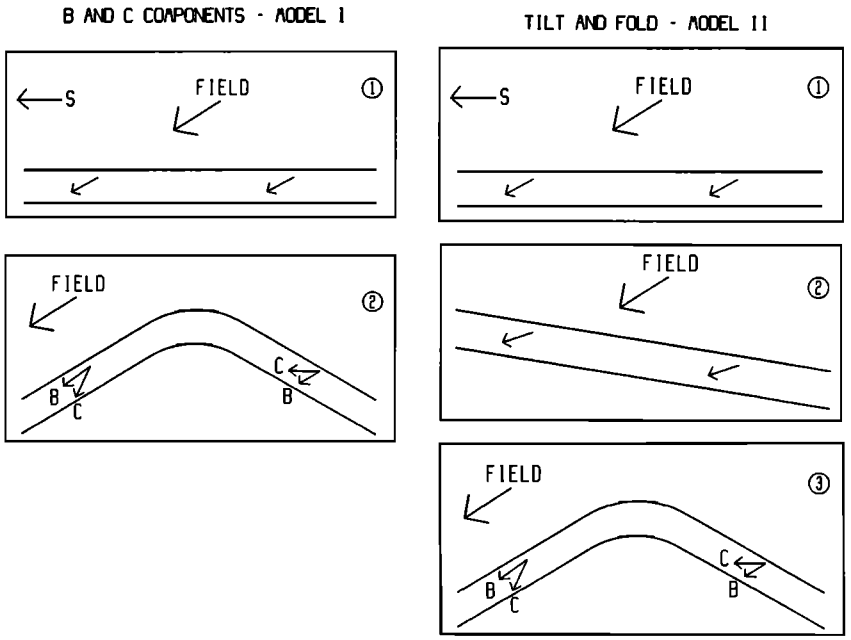



Fig. 13. Model I: C component acquired in flat lying beds at time 1; B component acquired during folding at time 2. Model II: C component acquired in flat lying beds at time 1: regional tilting at time 2 with no acquisition; the B component is acquired during localized folding at time 3.

21562202b, 1991, B3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/90JB02148 by Curtin University Library, Wiley Online Library on [14/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

acquired at this time, yielding the field relations discussed above. After this the beds are tilted farther to their present attitude. However, unfolding the beds approximately 70% yields an inclination which is apparently too steep in the present case, even though the north and south limb B vectors come into agreement. Another step may be called upon to explain the present results.

In Model II (Figure 13) time 1 is as before, the C component is acquired in flat lying beds. Then, the whole area is tilted to the north at time 2. No appreciable magnetization occurs at this time. Then at time 3, localized folding forms the Maringouin Anticline. The B component is acquired with a best north-south limb agreement at approximately 30% of folding. If the initial angle of tilt (model II, time 2) was 22° then correcting for 71% of tilt back to an original 22° paleoslope yields a B component mean of D=172.1°, I=5.1°, $\alpha_{95}$=4.4° for the upper 150 m of the Shepody Formation from the north and south limb of the Maringouin Anticline combined. The inclination (5.1°) is shallower than all of the C component mean inclinations calculated in this study and is consistent with a later acquisition of the B component during northward plate motion.

Any degree of initial tilting may be chosen for calculation purposes to match a preferred model, so without evidence for such a mechanism it remains merely a convenient explanation. Assymetric folding has also been suggested as a possible mechanism by which to explain the observed inclinations, but angles of tilt may also be chosen at will for this model to fit the desired outcome.

## Magnetostratigraphic Comparison

There are some uncertainties in the present extension of the Carboniferous magnetic stratigraphy. In spite of the difficulties in the Claremont Conglomerate, this part of the section can at least be characterized as dominantly normal polarity. The one reversed (R6) and two normal (N6 and N7) magnetozones defined could actually be a single normal polarity zone, or less likely, there could be multiple short reversed polarity zones. No normal zones have been ascribed to the upper Boss Point Formation or the Joggins Formation because there were no clearly normal samples and in no case were there two or more consecutive samples with trends toward normal polarity. The upper red facies of the Boss Point Formation appears to be largely remagnetized and the Joggins Formation may also be biased by the B component. In spite of this, the upper part of the section might reasonably be characterized as dominantly reversed with the possibility of short normal polarity zones, though they are not claimed here.

The present magnetic stratigraphy can be compared to the Paleozoic magnetic stratigraphy developed in the Soviet Union in the 1960s and 1970s by Khramov and others from the Russian Platform and adjacent areas [Khramov 1987] (Figure 14). The Soviet workers have correlated their magnetic stratigraphy to the Russian time scale and the two stratigraphies are here correlated according to the standard time scale of Harland et al. [1982] where the Bashkirian-Moscovian boundary is correlated with the Westphalian A-Westphalian B boundary and the Serpukhovian-Bashkirian boundary is correlated with the Namurian A-Namurian B boundary, which occurs within the Maringouin-Shepody sequence. The magnetic stratigraphy obtained in the Cumberland structural basin possesses considerably more detail than the Russian magnetic stratigraphy, nevertheless there is a first order similarity between the two stratigraphies. The long, lower Bashkirian normal zone could be correlative with N5 in the Shepody Formation. No sampling was done in the mid-Westphalian A in this study, so no comparison can be made with the long, mid-Bashkirian normal polarity zone. The upper Bashkirian and lower Moscovian short normal magnetozones are as yet unsupported by this study; however, short normal polarity zones in the upper Boss Point Formation and the Joggins Formation may be masked by overprinting. Overall, the Soviet magnetic stratigraphy contains roughly half normal polarity in the Namurian and into the Westphalian and dominantly reversed polarity later in the Westphalian. This gross estimation holds true for the Cumberland structural basin magnetic stratigraphy as well. However, the reversal frequency observed in the Canadian data indicates a much higher reversal frequency than would be determined from the Russian results. The reversals recorded in Canada cover an estimated 15- 20 m.y. interval yielding a reversal frequency approaching one reversal per million years. This rate is similar to the reversal frequency at the top of the "M" sequence anomalies just prior to the onset of the Cretaceous long normal superchron where the frequency is a little over one reversal per million years.

Roy and Morris [1983] suggested that the base of the PCRS should be placed within the Westphalian A based on results in the Cumberland structural basin. In the Russian Platform magnetic strati-

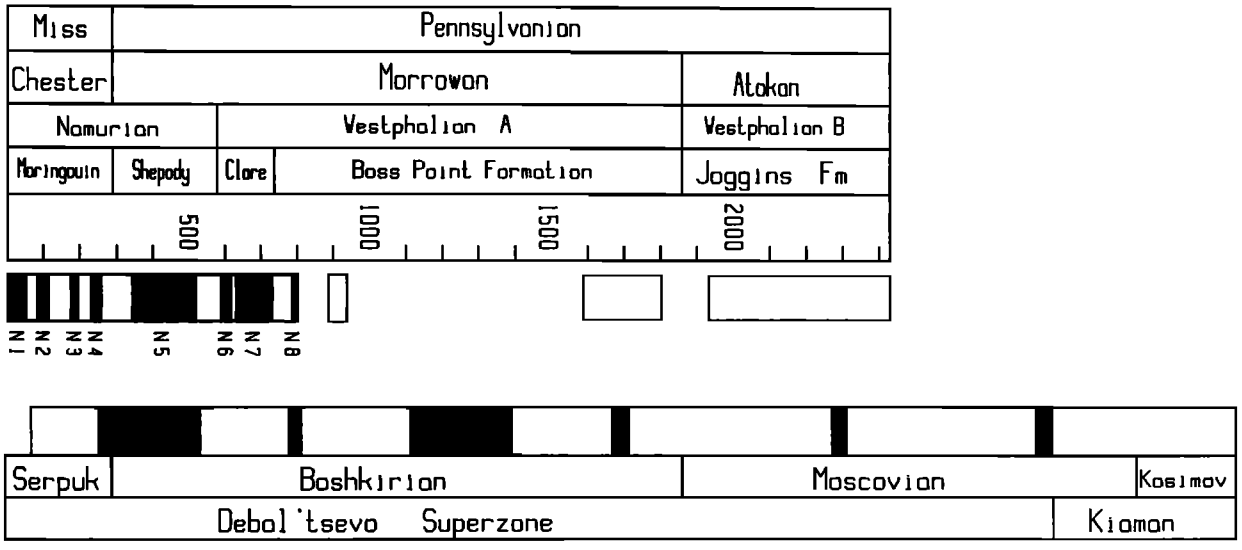



Fig. 14. Comparison of Cumberland structural basin results with the Russian platform results of Khramov and others.

21562202b 1991, B3 Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/90JB02148 by Curtin University Library, Wiley Online Library on [14/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

graphy the base of the Kiaman Hyperzone is placed after the last Debal'tsevo normal magnetozone in the upper Moscovian even though the polarity is dominantly reversed beginning in the upper Bashkirian and another short normal interval is defined within the Kiaman in the Asselian [Khramov and Rodionov, 1981]. The present results suggest dominantly mixed polarity in the Namurian and early Westphalian A and a change to dominantly reversed polarity by the late Westphalian A or early Westphalian B. However, the large sampling gap and overprinting problems make it difficult to further define the base of the PCRS.

## Conclusions

Eight normal (N1-N8) and eight reversed (R1-R8) polarity zones are defined in a composite stratigraphic sequence from Namurian A into Westphalian A. The present results bear a first-order agreement with the magnetic stratigraphy produced in the Soviet Union by Khramov and others; however, no correlation may be made in a detailed manner. The mean directions calculated for the units of the Nova Scotia section corroborate the shallowing of inclination observed in the Maringouin Peninsula section which could be documenting 5-7 cm/yr of northward plate motion during the Namurian and Westphalian.

Acknowledgements. The authors would like to express their appreciation to E.S. Belt, who kindly lent his dissertation. His help is much appreciated. The authors would also like to thank Sabine Musselwhite for her assistance in the second year of field work and Randy Enkin for the use of his great circle analysis software. This study was produced with funding from the National Science Foundation.

## References

Bell, W.A., Carboniferous rocks and fossil floras of northern Nova Scotia, *Mem. Geol. Surv. Can., 238*, 276 pp., 1944.

Belt, E.S., Revision of Nova Scotia middle Carboniferous units, *Am. J. Sci., 262*, 653-673, 1964.

Belt, E.S., Stratigraphy and paleogeography of Mabou Group and related middle Carboniferous facies, Nova Scotia, Canada, *Geol. Soc. Am. Bull., 76*, 777-801, 1965.

Bradley, D.C., Subsidence in late Paleozoic basins in the northern Appalachians, *Tectonics, 1*, 107-123, 1982.

Carroll, R.L., E.S. Belt, D.L. Dineley, D. Baird, and D.C. McGregor, *Twenty Fourth International Geological Congress, Field Excursion A59: Vertebrate Paleontology of Eastern Canada*, 113 pp., John D. McAra Limited, Calgary, Alberta, 1972.

Copeland, M.J., The arthropod fauna of the Upper Carboniferous rocks of the Maritime provinces, Mem. *Can. Geol. Surv., 286*, 69 pp., 1957.

Courtillot, V., and J. Besse, Magnetic field reversals, polar wander, and core-mantle coupling, *Science, 237*, 1140-1147, 1987.

DiVenere, V.J., and N.D. Opdyke, Paleomagnetism of the Maringouin and Shepody formations, New Brunswick: A Namurian magnetic stratigraphy, *Can. J. Earth Sci.*, 27, 803-810, 1990.

Fisher, R.A., Dispersion on a sphere, *Proc. R. Soc. London., Ser.A, 217*, 295-305, 1953.

Harland, W.B., A.V. Cox, P.G. Llewellyn, C.A.B. Pickton, A.G. Smith, and R.A. Walters, *A Geologic Time Scale*, 131 pp., Cambridge University Press, New York, 1982.

Irving, E., and G.A. Irving, Apparent polar wander paths Carboniferous through Cenozoic and the assembly of Gondwana, *Geophys. Surv., 5*, 141-188, 1982.

Irving, E., and G. Pullaiah, Reversals of the geomagnetic field, magnetostratigraphy, and relative magnitude of paleosecular variation in the Phanerozoic, *Earth Sci. Rev., 12*, 35-64, 1976.

Irving, E., and D.F. Strong, Paleomagnetism of the early Carboniferous Deer Lake Group, western Newfoundland: No evidence for Carboniferous displacement of "Acadia", *Earth Planet. Sci. Lett., 69*, 379-390, 1984.

Johnson, R.J.E., and R. Van der Voo, Dual-polarity Early Carboniferous remagnetization of the Fisset Brook Formation, Cape Breton Island, Nova Scotia, *Geophys. J., 97*, 259-273, 1989.

Kent, D.V., Further paleomagnetic evidence for oroclinal rotation in the central folded Appalachians from the Bloomsburg and the Mauch Chunk formations. *Tectonics, 7*, 749-759, 1988.

Kent, D.V., and N.D. Opdyke, Multicomponent magnetizations from the Mississippian Mauch Chunk Formation of the central Appalachians and their tectonic implications, *J. Geophys. Res., 90*, 5371-5383, 1985.

Khramov, A.N., *Paleomagnetology*, 308 pp., Springer-Verlag, New York, 1987.

Khramov, A.N., and V.P. Rodionov, The geomagnetic field during Palaeozoic time, in *Global Reconstruction and the Geomagnetic Field during the Palaeozoic*, edited by M.W. McElhinny, A.N. Khramov, M. Ozima, and D.A. Valencio, Adv. Earth Planet. Sci., 10, 99-115, 1981.

Kirschvink, J.L., The least-squares line and plane and the analysis of palaeomagnetic data, *Geophys. J. R. Astron. Soc., 62*, 699-718, 1980.

McCabe, C., M. Jackson, and B. Saffer, Regional patterns of magnetite authigenesis in the Appalachian Basin: Implications for the mechanism of Late Paleozoic remagnetization, *J. Geophys. Res., 94*, 10,429-10,443, 1989.

McFadden, P.L., and M.W. McElhinny, The combined analysis of remagnetization circles and direct observations in paleomagnetism, *Earth Planet. Sci. Lett., 87*, 161-172, 1988.

Nova Scotia Department of Mines and Energy, Cumberland Basin Project, map, *Map Sheet 21H/09-18, OFM 88-42*, Halifax, 1988.

Olivet, J.L., J. Bonnin, P. Beuzart, and J.M. Auzende, Cinématique de l'Atlantique Nord et Central, *Publ. Cent. Nat. Exploit. Océans, Rapp. Sci. Tech. Fr. 54*, 108 pp., 1984.

Roy, J.L., Paleomagnetism of the Cumberland Group and other Paleozoic formations, *Can. J. Earth Sci., 6*, 663-669, 1969.

Roy, J.L., and W.A. Morris, A review of paleomagnetic results from the Carboniferous of North America: The concept of Carboniferous geomagnetic field horizon markers, *Earth Planet. Sci. Lett., 65*, 167-181, 1983.

Roy, J.L., and J.K. Park, Paleomagnetism of the Hopewell Group, New Brunswick, *J. Geophys. Res., 74*, 594-604, 1969.

Scott, G.R., and C. Frohlich, Large-Volume, Magnetically Shielded Room: A New Design and Material, in *Magnetite Biomineralization and Magnetoreception in Organisms: A New Biomagnetism*, edited by J.L. Kirschvink, D.S. Jones, and B.J. MacFadden, pp. 197-220, Plenum, New York, 1985.

Stockmans, F., and Y. Willière, *Végétaux namuriens de la Belgique*, vol. 13, 382 pp., Association de l'Étude paléontologie et stratigraphie houillères, 1953.

Symons, D.T.A., Early Permian pole: Evidence from the Pictou red beds, Prince Edward Island, Canada, *Geology, 18*, 234-237, 1990.

Torsvik, T.H., O. Lyse, G. Atlerås, and B.J. Buck, Paleozoic paleomagnetic results from Scotland and their bearing on the British apparent polar wander path, *Phys. Earth Planet. Inter., 55*, 93-105, 1989.

V. J. DiVenere, Lamont Doherty Geological Observatory, Palisades, NY 10964.

N. D. Opdyke, Department of Geology, 1112 Turlington, University of Florida, Gainesville, FL, 32611.

(Received August 3, 1990;
revised September 27, 1990;
accepted September 28, 1990.)

21562202b, 1991, B3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/90JB02148 by Curtin University Library, Wiley Online Library on [14/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License